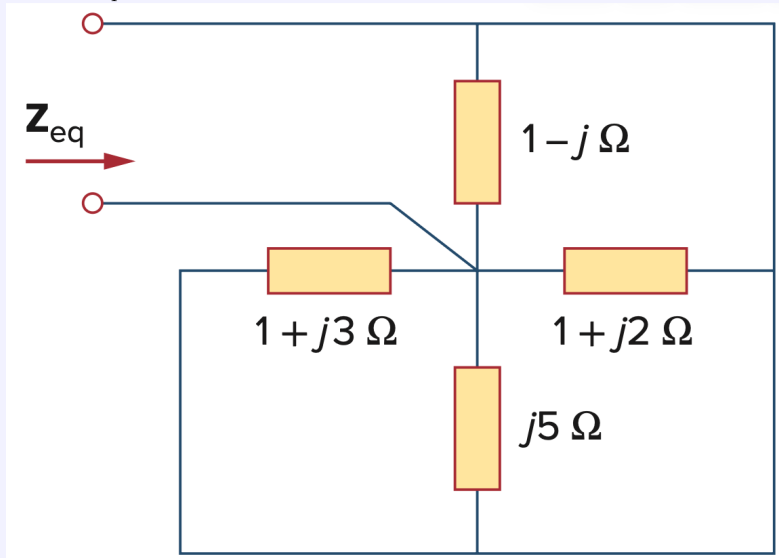


Exercise Sheet #7

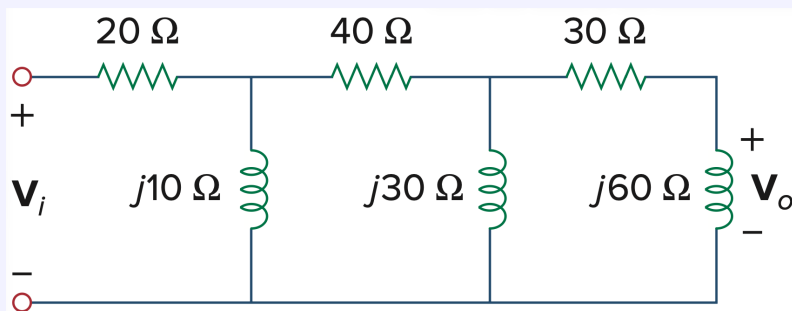
AC 2

Exercise 1: Equivalent Impedance

Find $\underline{Z}_{eq}$ in the circuit below.

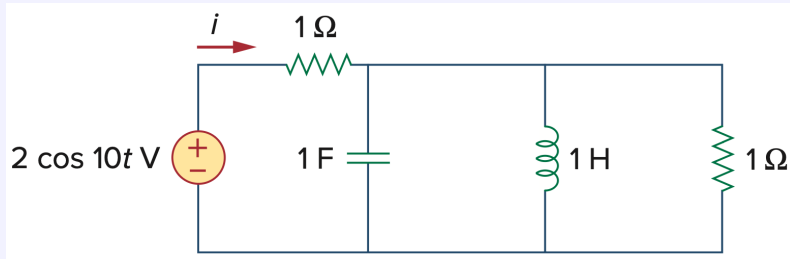
Exercise 2: Stepwise Voltage Division

- Calculate the phase shift of the circuit below.
- State whether the phase shift is leading or lagging (output with respect to input).
- Determine the magnitude of the output when the input is 120 V.



Exercise 3: Nodal Analysis

Determine i in the circuit below.



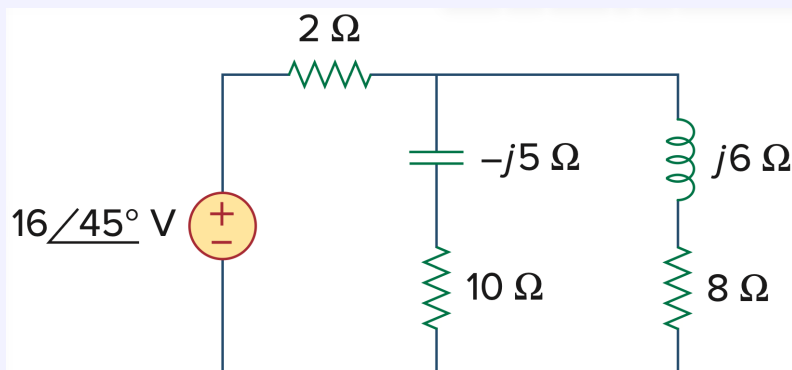
Exercise 4: Instantaneous and Average Power

If $v(t) = 160 \cos(50t) \text{ V}$ and $i(t) = -33 \sin(50t - 30^\circ) \text{ A}$, calculate the instantaneous power and the average power.

Exercise 5: Complex Power Calculation

For the entire circuit below, calculate:

- the complex power
- the average power delivered by the source
- the reactive power
- the apparent power
- the power factor



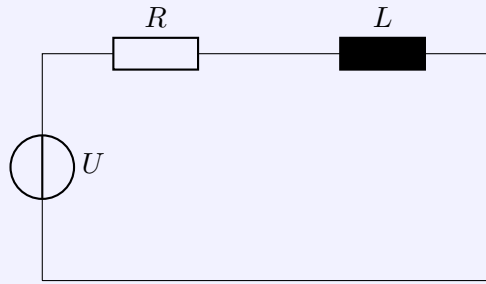
Exercise 6: RMS Values and Average Power

The voltage across a load and the current through it are given by $v(t) = 20 + 60 \cos(100t) \text{ V}$ and $i(t) = 1 - 0.5 \sin(100t) \text{ A}$. Find:

- the RMS values of the voltage and of the current
- the average power dissipated in the load

Exercise 7: Complex Network Analysis

Given the following network with a sinusoidal voltage source $U = 230 \text{ V}$, $f = 50 \text{ Hz}$ and the components $R = 20 \Omega$, $L = 100 \text{ mH}$:



Determine:

1. the complex impedance Z ,
2. the RMS current I ,
3. the phase shift between current and voltage,
4. the power factor $\cos \varphi$,
5. the active power P ,
6. the reactive power Q .

Exercise 8: Reactive Power Compensation

An AC motor with impedance $\underline{Z} = 4.2 + j 3.6 \Omega$ is supplied by a 220-V, 60-Hz source.

- a) Find the power factor, the real power P , and the reactive power Q .
- b) Determine the capacitor required to be connected in parallel with the motor so that the power factor is corrected to unity.

Exercise 9: Series Resonance

A series RLC network has $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, and $C = 1 \mu\text{F}$. Calculate the impedance at resonance and at one-fourth, one-half, twice, and four times the resonant frequency.

Exercise 10: Parallel Resonance

Rework Exercise 6 if the elements are connected in parallel.